MODULE 3 VESSEL DISORDERS AND FETAL CIRCULATION

Blood Vessel Disorders and Fetal Circulation.

BLOCK 3 · MODULE 5: DISORDERS AND THE FETAL PLAN

This chapter covers two topics. First, the common disorders of blood vessels, which mostly narrow or weaken the vessel wall. Second, fetal circulation, the rearranged plan a fetus uses before its lungs work, and what each detour leaves behind after birth.

BY THE END

1. Describe atherosclerosis and how it changes an artery wall.
2. Name common arterial and venous disorders.
3. Explain how fetal circulation differs from the adult plan.
4. Name the fetal shunts and the adult remnant each one becomes.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Blood vessels

drsrennie-stack.github.io/new-build-bio4-solano/blood-vessels-concept-videos.html

An Overview

THE TWO HALVES OF THIS CHAPTER

Structure	What it is
Healthy vessel	A vessel with a smooth endothelial lining and a wall that can stretch and recoil
Vascular disorder	A disease of a blood vessel; most disorders narrow the lumen, weaken the wall, or let blood pool
Fetal circulation	The circulatory plan of a fetus, which gets its oxygen from the placenta rather than from its own lungs
Shunt	A special vessel or opening that routes blood around an organ; the fetus uses shunts to bypass the lungs and liver

Atherosclerosis

Atherosclerosis is the most important vessel disease, the root of heart attacks and strokes. It is a slow, structural change in the artery wall.

THE DISEASE, STRUCTURE BY STRUCTURE

Structure	What it is
Atherosclerosis	A disease in which fatty plaques build up within the walls of arteries
Endothelial injury	Damage to the smooth tunica intima lining, the starting point of a plaque
Atheroma	Also called a plaque, a deposit of fatty material, cells, and debris within the artery wall
Narrowed lumen	As a plaque grows it bulges inward, narrowing the channel and reducing blood flow
Arteriosclerosis	The general stiffening and thickening of artery walls; atherosclerosis is its most common form
Thrombus	A clot that can form on a roughened plaque and suddenly block the vessel
Embolus	A clot or fragment that breaks free and travels through the bloodstream to lodge elsewhere

Arterial Disorders

Arteries carry blood under high pressure, so their disorders tend to narrow the channel or weaken the wall.

COMPARE THE COMMON ONES

Disorder	What it is	Why it matters
Aneurysm	A balloon-like bulge in a weakened artery wall	It can rupture; common in the aorta and the arteries of the brain
Atherosclerosis	Fatty plaque buildup that narrows an artery	The leading cause of coronary artery disease and stroke
Coronary artery disease	Narrowing of the coronary arteries that supply the heart wall	Can starve the myocardium and cause a heart attack
Peripheral artery disease	Narrowing of the arteries of the limbs, usually the legs	Causes pain on walking as the muscles are starved of blood
Hypertension	Chronically high pressure within the arteries	Strains the heart and damages vessel walls over time
Stroke	Loss of blood supply to part of the brain, from a blocked or burst artery	Brain tissue begins to die within minutes

Venous Disorders

Veins carry blood under low pressure, so their disorders tend to involve failed valves and pooled or clotted blood.

COMPARE THE COMMON ONES

Disorder	What it is	Why it matters
Varicose veins	Superficial veins that become twisted and swollen when their valves fail	Blood pools and the veins bulge visibly, most often in the legs
Deep vein thrombosis	A blood clot in a deep vein, usually of the leg	The clot can break free and travel through the bloodstream
Pulmonary embolism	A clot, often from a deep vein, that lodges in an artery of the lung	A life-threatening emergency that blocks blood flow to the lung
Phlebitis	Inflammation of a vein	Causes pain, redness, and tenderness along the vein
Hemorrhoids	Varicose veins of the rectal and anal region	A common cause of rectal discomfort and bleeding
Chronic venous insufficiency	Long-term failure of the valves of the leg veins	Leads to persistent swelling and skin changes in the lower leg

Fetal Circulation, an Overview

Before birth, a fetus does not breathe air or digest food, so its circulation is rearranged. Oxygen and nutrients come from the placenta, and blood is routed around the organs that are not yet in use.

WHAT IS DIFFERENT BEFORE BIRTH

Structure	What it is
The placenta	The organ where fetal blood picks up oxygen and nutrients from the mother and gives up wastes
Umbilical cord	The cord connecting the fetus to the placenta, carrying the umbilical vessels
The lungs are bypassed	The fetal lungs are collapsed and not used for gas exchange, so most blood is routed around them
The liver is largely bypassed	Much fetal blood skips the liver, since the placenta handles that filtering
Three shunts	The ductus venosus, the foramen ovale, and the ductus arteriosus, the detours that route blood around the liver and lungs

The Fetal Vessels and Shunts

Five special structures carry and reroute fetal blood. Compare what each one is and what it does.

THE FETAL STRUCTURES

Structure	What it is	What it does
Umbilical vein	A vein in the umbilical cord	Carries oxygen-rich blood from the placenta to the fetus
Ductus venosus	A shunt passing through the fetal liver	Lets most blood bypass the liver and flow straight toward the inferior vena cava
Foramen ovale	An opening in the interatrial septum	Lets blood pass directly from the right atrium to the left atrium, bypassing the lungs
Ductus arteriosus	A short vessel between the pulmonary trunk and the aorta	Lets blood bypass the lungs by shunting it from the pulmonary trunk into the aorta
Umbilical arteries	Two arteries in the umbilical cord	Carry oxygen-poor blood from the fetus back to the placenta

The Shunts After Birth

At birth the lungs inflate and the placenta is cut off. Each fetal shunt is no longer needed, closes, and is left behind as a small ligament or remnant.

COMPARE EACH ONE WITH WHAT IT BECOMES

Fetal structure	Adult remnant	Note
Umbilical vein	The ligamentum teres, the round ligament of the liver	Runs in the free edge of the falciform ligament
Ductus venosus	The ligamentum venosum	A fibrous cord on the inferior surface of the liver
Foramen ovale	The fossa ovalis	A shallow oval depression in the interatrial septum
Ductus arteriosus	The ligamentum arteriosum	A short cord between the pulmonary trunk and the aorta
Umbilical arteries	The medial umbilical ligaments	Paired cords on the inner surface of the anterior abdominal wall

KEY POINT

Every fetal shunt becomes a ligament, and the name usually keeps a piece of the original. Ductus venosus becomes ligamentum venosum, ductus arteriosus becomes ligamentum arteriosum. The one that breaks the pattern is the foramen ovale, which becomes a depression rather than a cord, because it was a hole and not a vessel.

WHEN THE DETOUR DOES NOT CLOSE

The fetal shunts are meant to close in the first hours and weeks after birth. When the **ductus arteriosus** stays open, the result is a patent ductus arteriosus; when the **foramen ovale** stays open, a patent

foramen ovale. Both leave an abnormal connection between the two sides of the circulation, and both are named for the fetal structure that failed to seal.

Atherosclerosis ties the disorders together. A plaque that narrows a coronary artery causes a heart attack; one that narrows or bursts a brain artery causes a stroke; one in the leg causes peripheral artery disease. The same wall change, in different vessels, names different diseases.

A **deep vein thrombosis** in the leg is dangerous for where the clot can go. If it breaks free, it travels up the venae cavae, through the right heart, and lodges in a lung artery as a **pulmonary embolism**. The route the clot takes is just the normal venous return.